



July 13, 2014

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Re: Statistical Analysis of Sales Model

RESEARCH FIELD: Business Economics

PROJECT TITLE: Results of Sales Prediction Model

1.0 - PROJECT DESCRIPTION

Forecasting is a frequently used tool for making short-term and long-term decisions by using historical data to produce estimates for future developments such as profits, expenditures and sales. Particularly in business, forecasting plays a very important role in helping managers prepare for the future in a constantly changing economy.

The goal of this project is to use historical data to predict future sales using a statistical prediction model. The historical data contains observed quarterly values of wage and salary disbursements, employment, building material dealer sales, auto dealer sales, furniture dealer sales and general merchandise dealer sales. These quarterly values span from year 1979 to year 1989. The values to predict are sales for the four quarters in the year 1990. The predicted sales are then compared to the actual sales. The accuracy of the model is measured by determining whether actual sales are within the 95% prediction interval obtained using the predicted sales.

2.0 - DATA

These data are published monthly by the Bureau of Economic Analysis in the statistical section of the Survey of Current Business publication. The data consists of 7 variables and 44 observations. There is one observation of each variable for each quarter starting from the first quarter of 1979 to the last quarter of 1990 (11 years x 4 quarters = 44).

Variable Names:

1. TIME: Quarter, numbered 1 to 44, from 1st quarter 1979 to 4th quarter 1989
2. WASA: National income wage and salary disbursements (\$ billions)
3. EMPL: Employees on payrolls of non-agricultural establishments (thousands)
4. BLDG: Building material dealer sales (\$ millions)
5. AUTO: Automotive dealer sales (\$ millions)

6. FURN: Furniture and home furnishings dealer sales (\$ millions)
7. GMER: General merchandise dealer sales (\$ millions)

The objective of the prediction model is to predict the dealer sales (for each building material, auto, furniture and general merchandise) that will occur in the next four quarters (i.e. year 1990). Accordingly, the response variables in the model are BLDG, AUTO, FURN and GMER. The predictor variables are TIME, WASA and EMPL. Lag (past) values of sales can also be used as predictors. All of the variables are quantitative. Some basic characteristics of the variables can be found in Table 2.1.

Table 2.1 Descriptive Statistics of WASA, EMPL, BLDG, AUTO, FURN and GMER

Variable	Mean	StDev	Minimum	Median	Maximum	Range	IQR
WASA	1,856	428	1,193	1,806	2,613	1,419	683
EMPL	96,348	6,781	87,973	94,639	110,260	22,287	11,764
BLDG	17,082	4,744	9,392	16,677	26,344	16,952	7,603
AUTO	66,074	21,057	38,703	68,052	102,934	64,231	39,573
FURN	16,171	5,040	9,203	15,485	27,486	18,283	8,982
GMER	37,891	11,104	21,909	36,407	69,966	48,057	13,237

3.0 - EXPLORATORY DATA ANALYSIS (EDA)

3.1 - Data Visualization

The distribution of each variable is displayed in the boxplots contained in Figures 3.1a – 3.1c. The box represents the interquartile range (i.e. mid-range values) of the variable and is divided by a horizontal line which represents the median value. Each box has whiskers that extend outward to the minimum and maximum value of the variable. If existing, outliers are represented by asterisks. An outlier is a data value that is noticeably distant from the other values. Unfortunately, outliers can potentially reduce the reliability of a statistical model.

Figure 3.1a Boxplot of WASA

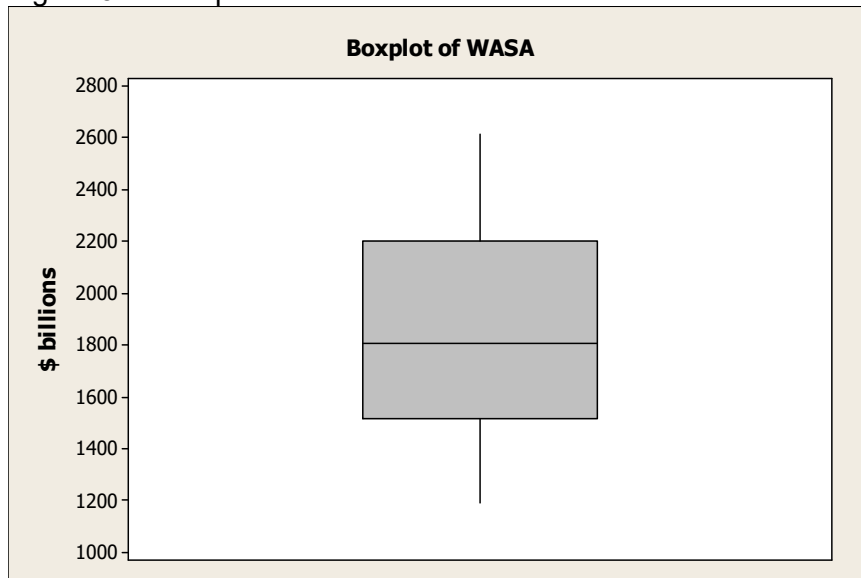


Figure 3.1b Boxplot of EMPL

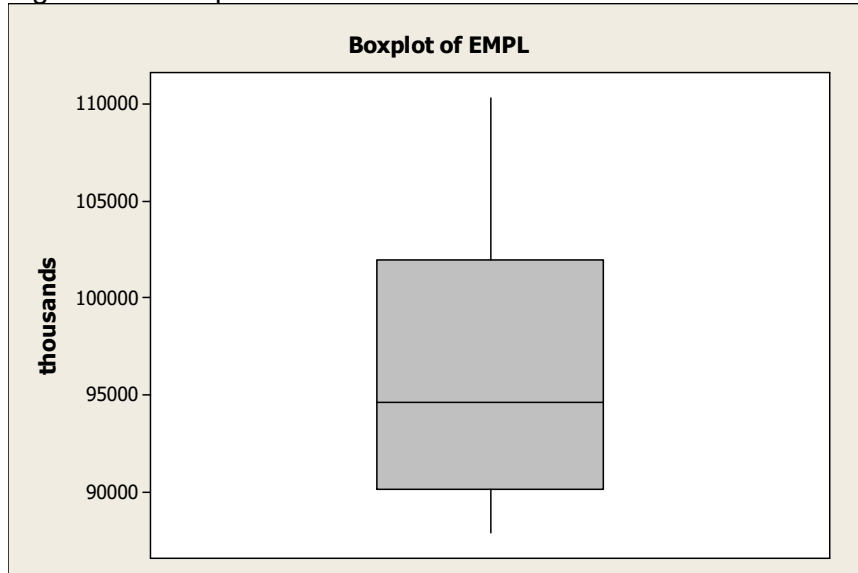


Figure 3.1c Boxplot of BLDG, AUTO, FURN and GMER

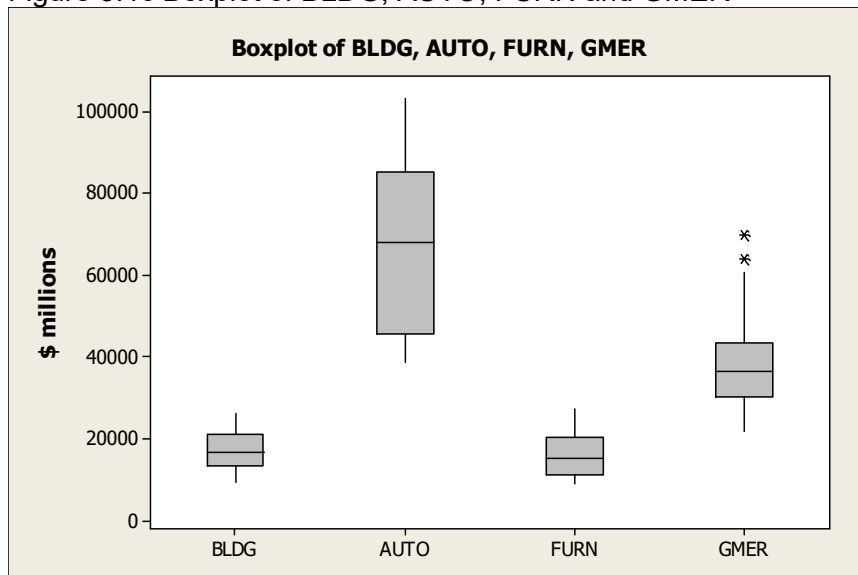
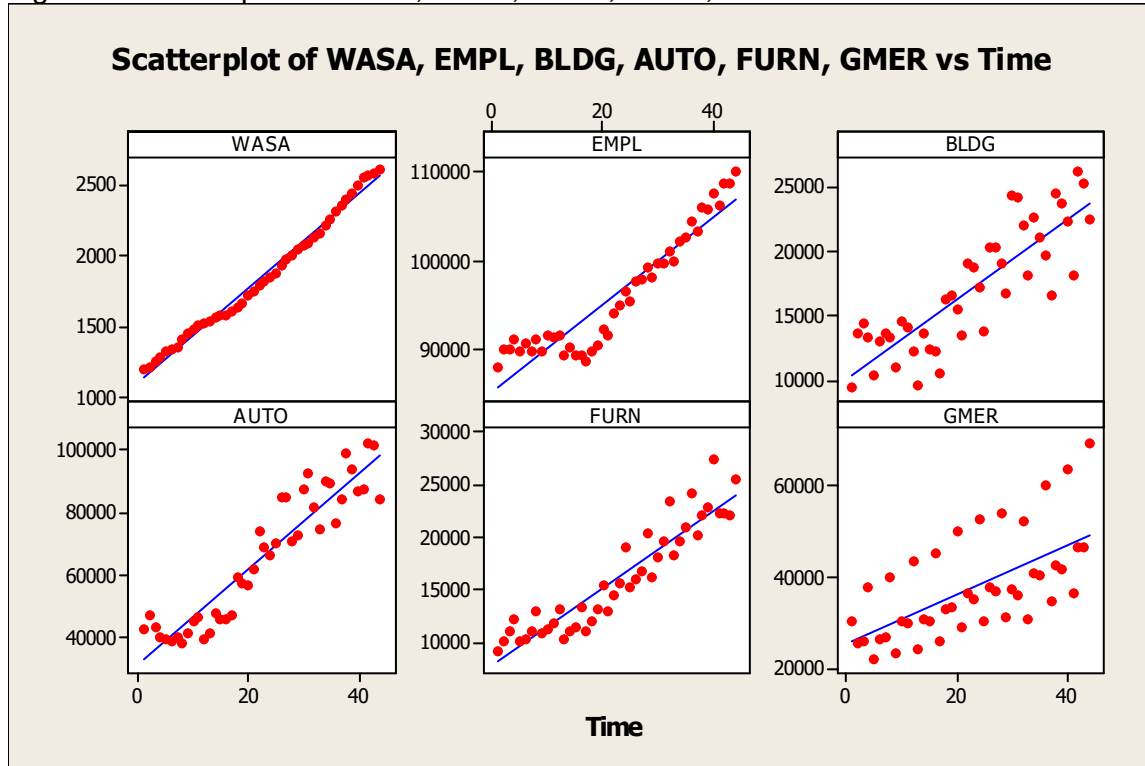


Figure 3.1c compares the distributions among the dealer sales. It is clear that automotive dealer sales has a much wider spread of values than the other sales. Building and furniture dealer sales have a similar range of sales. General merchandise dealer sales have two outliers. Since the outliers don't appear to be flagrantly distant and removing them will reduce the number of observations available for statistical analysis, they will remain in the data.

3.2 – Examining Linear Relationships

In addition to the distribution of the variables, whether or not the variables have a linear relationship with TIME is also of interest since it will influence what is done with the data. The scatter plots in Figure 3.2 show that the variables have a linear relationship with TIME.

Figure 3.2 Scatterplot of WASA, EMPL, BLDG, AUTO, FURN and GMER versus Time



All the variables appear to have a strong positive linear relationship with TIME. This means that the variables increase as time goes on. WASA appears to have the strongest relationship and GMER has the weakest. It is also possible that the variables are strongly linearly correlated with each other (multicollinearity). The presence of multicollinearity is problematic because it can cause the variable coefficient estimates to change erratically when a small change to the model or the data is made. This makes the model unstable. Multicollinearity will be assessed after the final form of the data series has been selected.

3.3 – Time Series Analysis Data Preparation

As stated earlier, each variable is observed quarterly -- at equally spaced successive points in time. Thus, the pairing of each variable with TIME creates a times series. So, the EDA includes creating a time series plot of each variable and noting the characteristics of the series. The time series model assumptions require that the time series be stationary. Stationary means that the time series has neither trend nor non-constant variance. Since the data is quarterly, the time series are expected to have seasonality. Seasonality occurs when the time series exhibits a pattern that repeats every period (which is every four quarters in this case). The time series are also expected to have an upward trend (as seen in the scatter plots of Figure 3.2).

Figure 3.3a WASA – The time series plot shows trend since the values steadily increase. The values seem to dip slightly every second quarter, so there may also be seasonality as well.

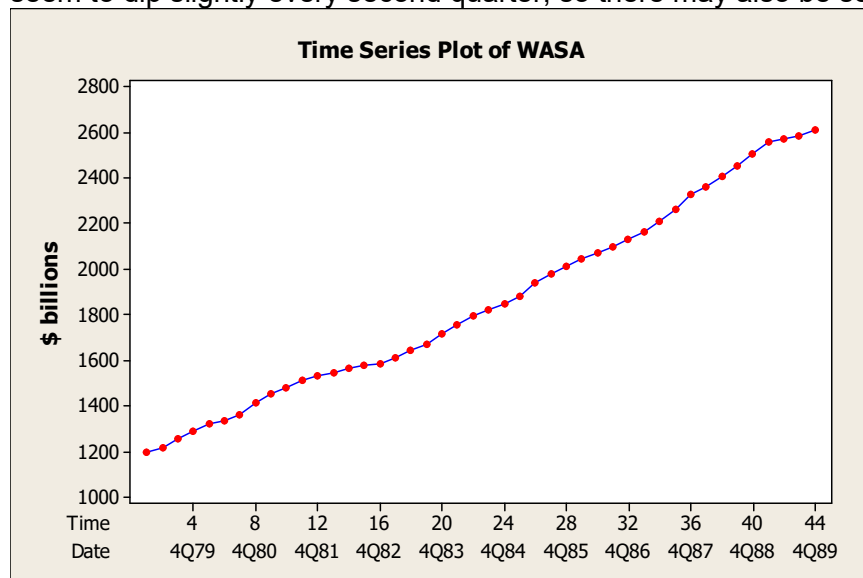


Figure 3.3b EMPL – The time series plot seems stationary until the first quarter of 1983, then the values trend upward. The values appear to have a pattern, so seasonality may be present.

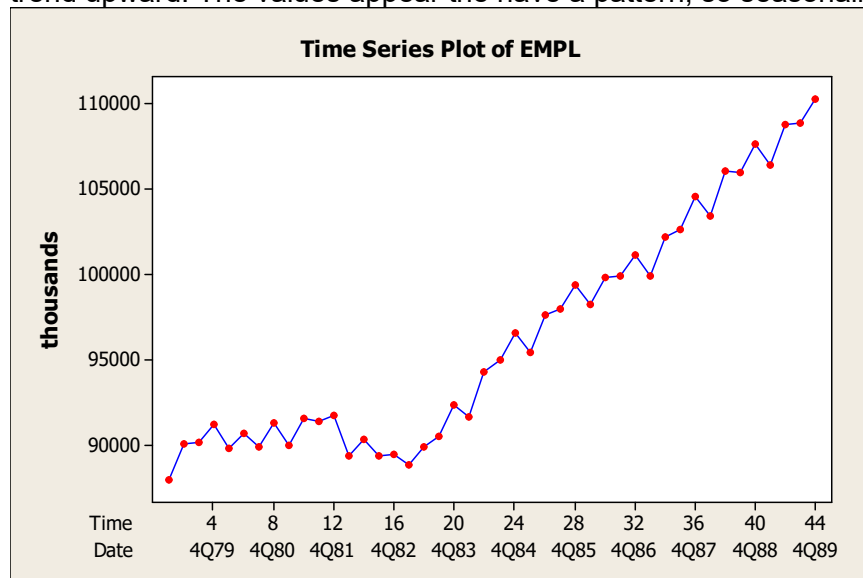


Figure 3.3c BLDG –The time series plot seems stationary until the first quarter of 1983, then sales trend upward. The values appear the have a pattern that repeats every four quarters, so seasonality may be present.

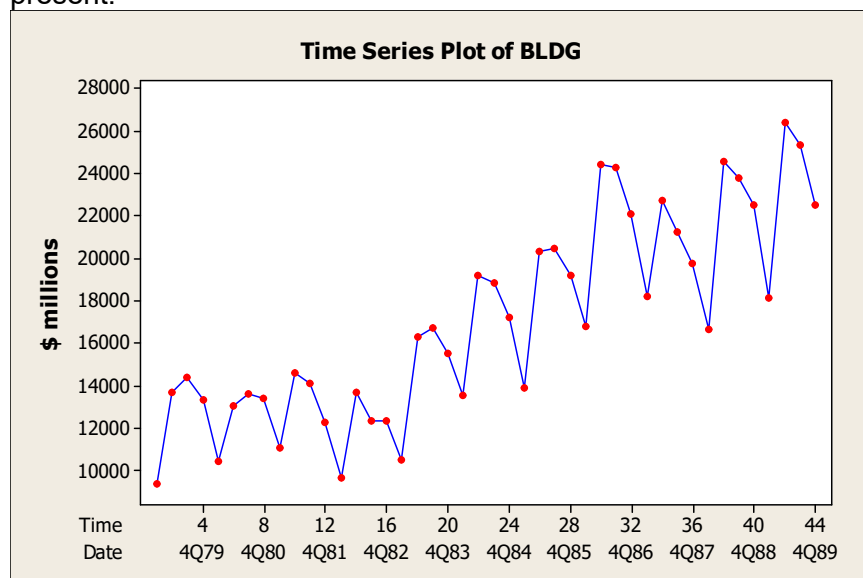


Figure 3.3d AUTO –The time series plot seems stationary until the first quarter of 1983, then sales trend upward. The values appear the have a pattern that repeats every four quarters, so seasonality may be present. The variance of sales increases with time.

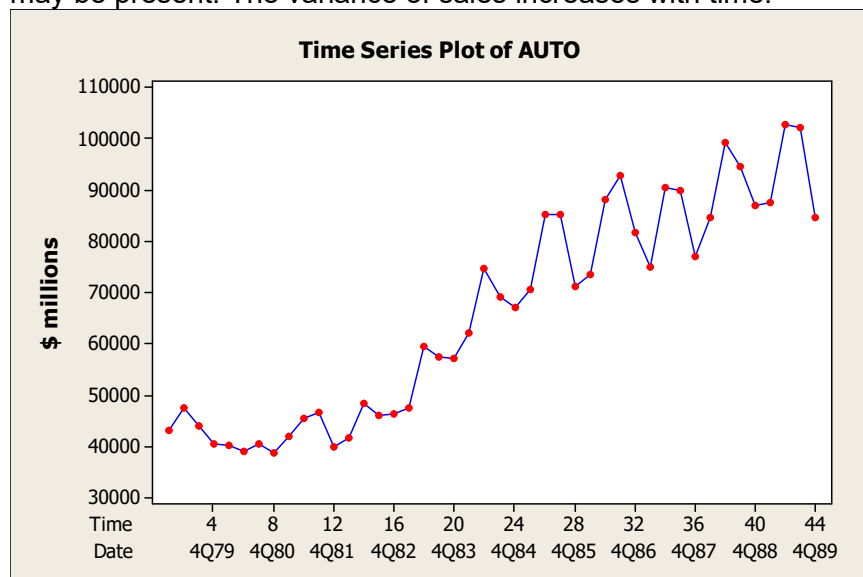


Figure 3.3e FURN –The time series plot shows that sales generally trend upward. Sales appear to have a pattern that repeats every four quarters, so seasonality may be present. The variance of sales increases with time.

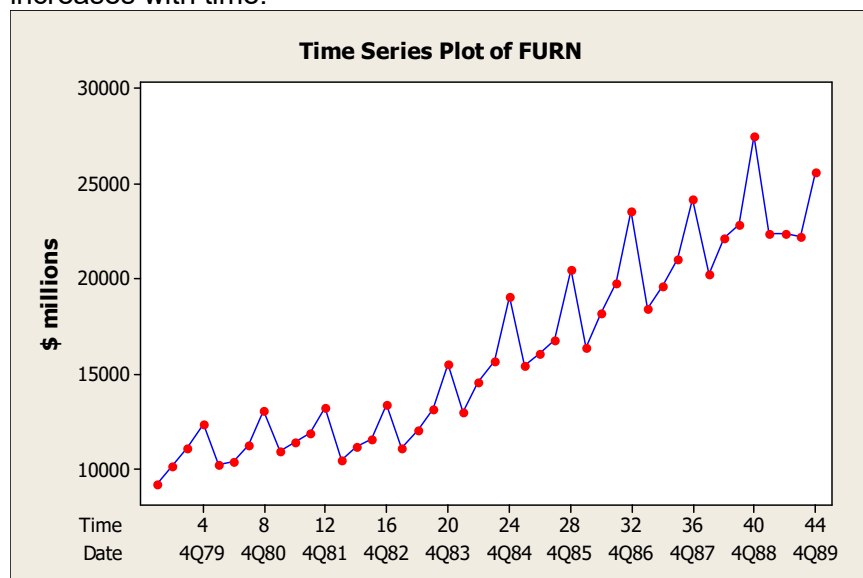
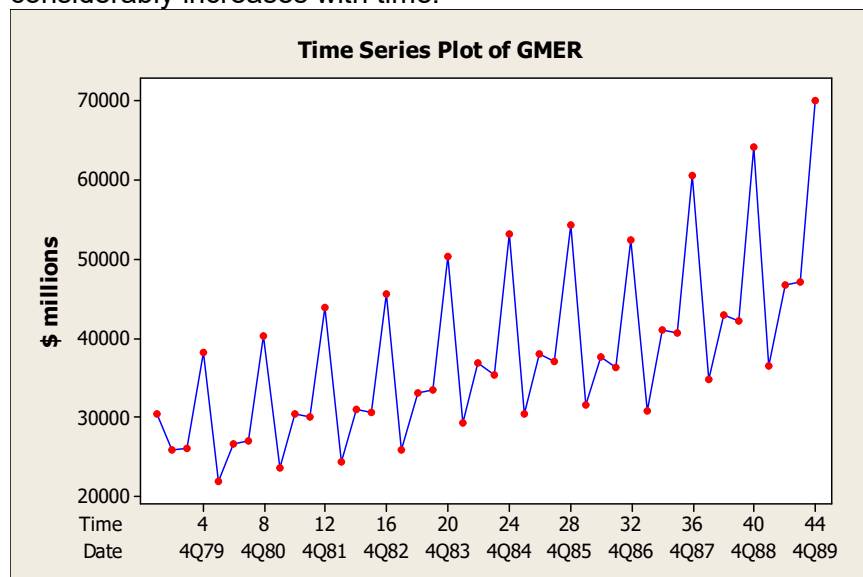


Figure 3.3f GMER –The time series plot shows that sales generally trend upward. Sales appear to have a strong pattern that repeats every four quarters, so seasonality is likely present. The variance of sales considerably increases with time.



All of the plots in Figures 3.3a - 3.3f showed trend, so each series was detrended using linear regression. The detrending process was performed by regressing each variable on time and then using the regression residuals instead of the original data series. Next, a fourth difference was applied to the detrended series to account for seasonality that repeats every four quarters.

Now that each time series is stationary, they can be fit to an appropriate time series model. Figures 3.4a - 3.4f illustrate the detrended, stationary time series.

Figure 3.4a Time series plot of DTWASA

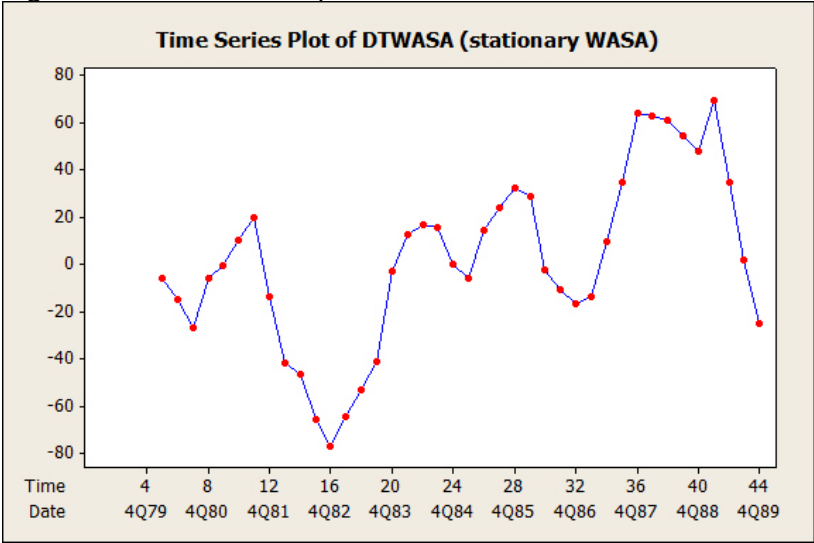


Figure 3.4b Time series plot of DTEML

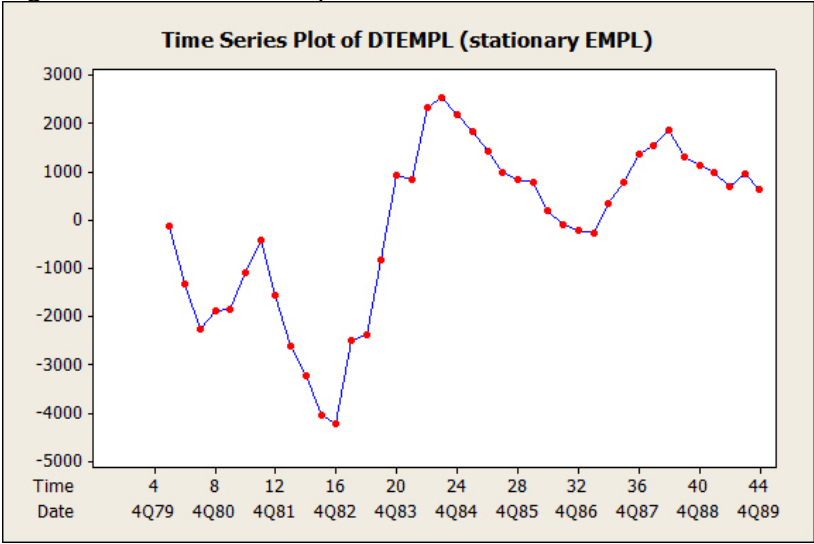


Figure 3.4c Time series plot of DTBLDG

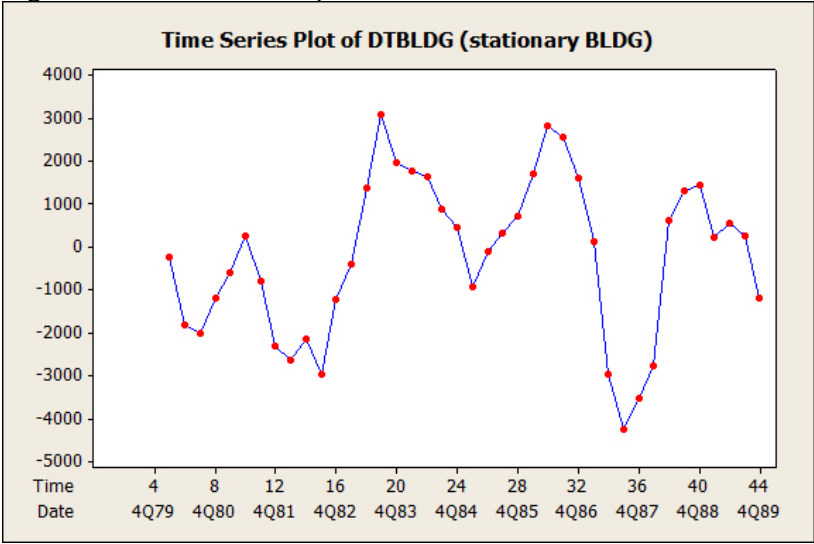


Figure 3.4d Time series plot of DTAUTO

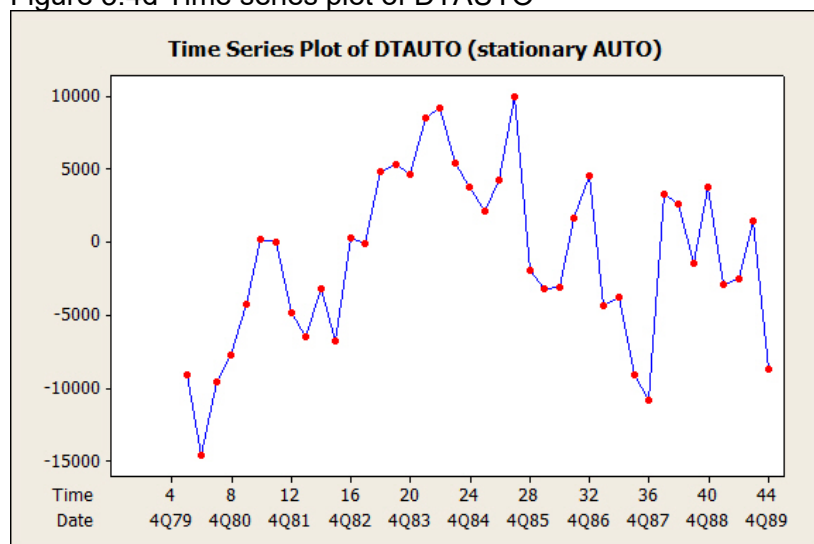


Figure 3.4e Time series plot of DTFURN

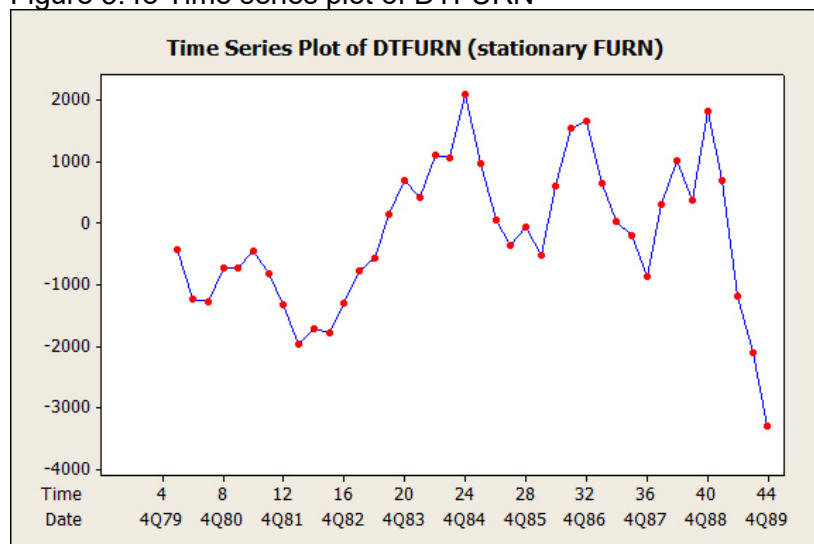
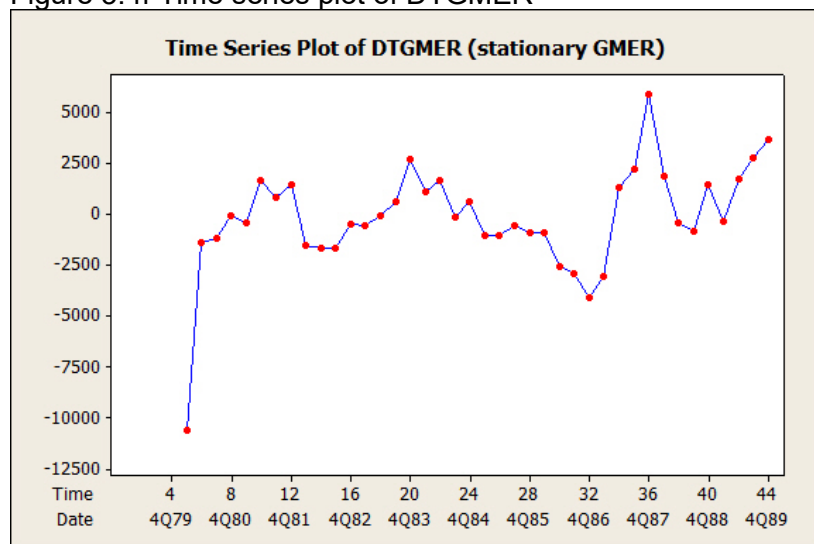


Figure 3.4f Time series plot of DTGMER



Further, the new stationary series can be tested for multicollinearity by using variance inflation factors (VIF). A VIF measures the severity of multicollinearity in a regression model. VIFs smaller than 5

indicate moderate multicollinearity. Table 3.3 lists the VIFs for each times series when regressed on all the other time series. The largest VIF is 3.6, thus multicollinearity is not a concern.

Table 3.3 Variance inflation factors for each multiple regression model among the time series

	Predictors					
	dtwasa	dtempl	dtbldg	dtauto	dtfurn	dtgmer
Response	dtwasa	1.8	2.0	2.1	2.3	1.5
	dtempl	1.5	2.0	2.3	2.3	1.5
	dtbldg	3.1	3.6	1.8	2.4	1.4
	dtauto	2.9	3.6	1.6	2.1	1.3
	dtfurn	3.0	3.5	2.0	2.0	1.3
	dtgmer	3.0	3.6	1.8	2.0	2.0

4.0 –STATISTICAL ANALYSIS

The time series were fit to a p-lag vector autoregressive (VAR(p)) model using the ar.ols function of the R 3.1 statistical software. The model allows use of a multivariate time series. The form is such that each variable is a linear function of past values of itself and past values of the other variables. The ar.ols function uses ordinary least squares to produce a separate linear model for each time series. The order of the model is determined by the number of lags needed to obtain a reasonable fit. The model selected is a VAR(3) model. The number three indicates that three lag values are used in the model. The form of the VAR(3) model for each of the n = 6 time series (DTWASA, DTEMPL, DTBLDG, DTAUTO, DTFURN and DTGMER) can be written as follows:

$$x_{t,1} = \alpha_1 + \varphi_{11}x_{t-1,1} + \varphi_{12}x_{t-1,2} + \varphi_{13}x_{t-1,3} + \varphi_{14}x_{t-1,4} + \varphi_{15}x_{t-1,5} + \varphi_{16}x_{t-1,6} + \varphi_{11}x_{t-2,1} + \varphi_{12}x_{t-2,2} + \varphi_{13}x_{t-2,3} + \varphi_{14}x_{t-2,4} + \varphi_{15}x_{t-2,5} + \varphi_{16}x_{t-2,6} + \varphi_{11}x_{t-3,1} + \varphi_{12}x_{t-3,2} + \varphi_{13}x_{t-3,3} + \varphi_{14}x_{t-3,4} + \varphi_{15}x_{t-3,5} + \varphi_{16}x_{t-3,6} + \omega_{t,1}$$

$$x_{t,2} = \alpha_1 + \varphi_{21}x_{t-1,1} + \varphi_{22}x_{t-1,2} + \varphi_{23}x_{t-1,3} + \varphi_{24}x_{t-1,4} + \varphi_{25}x_{t-1,5} + \varphi_{26}x_{t-1,6} + \varphi_{21}x_{t-2,1} + \varphi_{22}x_{t-2,2} + \varphi_{23}x_{t-2,3} + \varphi_{24}x_{t-2,4} + \varphi_{25}x_{t-2,5} + \varphi_{26}x_{t-2,6} + \varphi_{21}x_{t-3,1} + \varphi_{22}x_{t-3,2} + \varphi_{23}x_{t-3,3} + \varphi_{24}x_{t-3,4} + \varphi_{25}x_{t-3,5} + \varphi_{26}x_{t-3,6} + \omega_{t,2}$$

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$$x_{t,6} = \alpha_1 + \varphi_{61}x_{t-1,1} + \varphi_{62}x_{t-1,2} + \varphi_{63}x_{t-1,3} + \varphi_{64}x_{t-1,4} + \varphi_{65}x_{t-1,5} + \varphi_{66}x_{t-1,6} + \varphi_{11}x_{t-2,1} + \varphi_{12}x_{t-2,2} + \varphi_{13}x_{t-2,3} + \varphi_{14}x_{t-2,4} + \varphi_{15}x_{t-2,5} + \varphi_{16}x_{t-2,6} + \varphi_{11}x_{t-3,1} + \varphi_{12}x_{t-3,2} + \varphi_{13}x_{t-3,3} + \varphi_{14}x_{t-3,4} + \varphi_{15}x_{t-3,5} + \varphi_{16}x_{t-3,6} + \omega_{t,6}$$

Where x is the time series, α is the intercept, φ is the coefficient estimate for each lag value and ω is the uncorrelated error term. The ar.ols function returns a model for every time series used in the function even though we are only interested in modelling four of the time series (DTBLDG, DTAUTO, DTFURN and DTGMER).

The appropriateness of the model is checked by examining the autocorrelation function (ACF) plot of the residuals of each variable in the model. An ACF plot with no significant lags is desirable since it indicates that the residuals are uncorrelated. The ACF plots (see Figures 4.1a – 4.1f) of the VAR(3) model were acceptable.

Figure 4.1a ACF for DTWASA

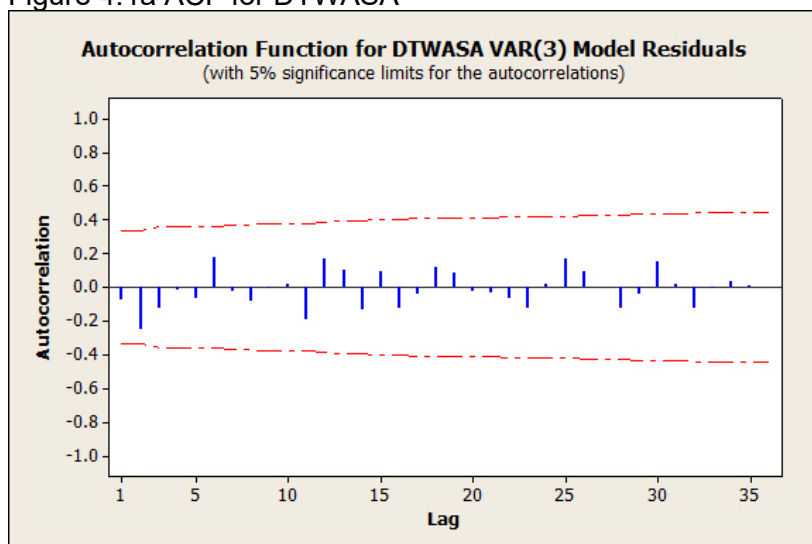


Figure 4.1b ACF for DTEMPL

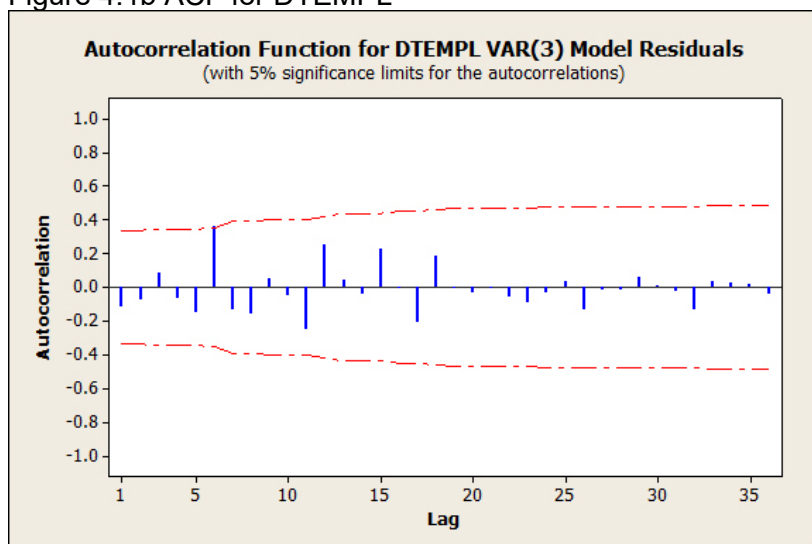


Figure 4.1c ACF for DTBLDG

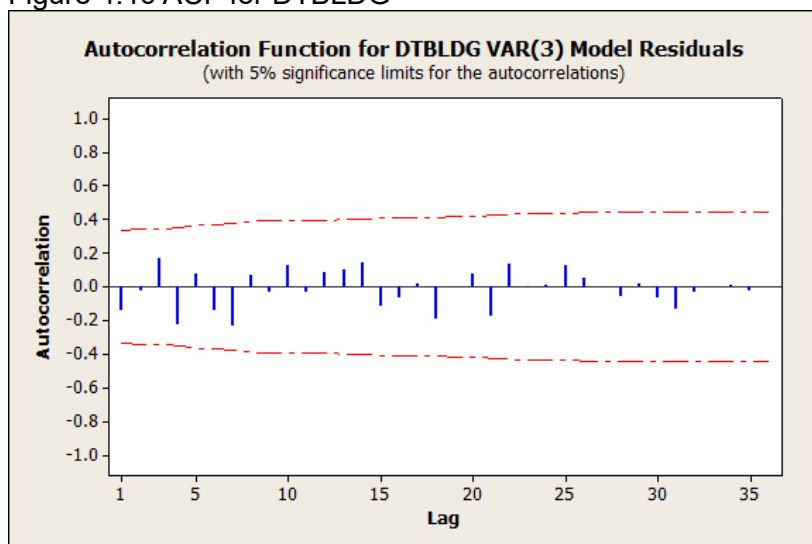


Figure 4.1d ACF for DTAUTO

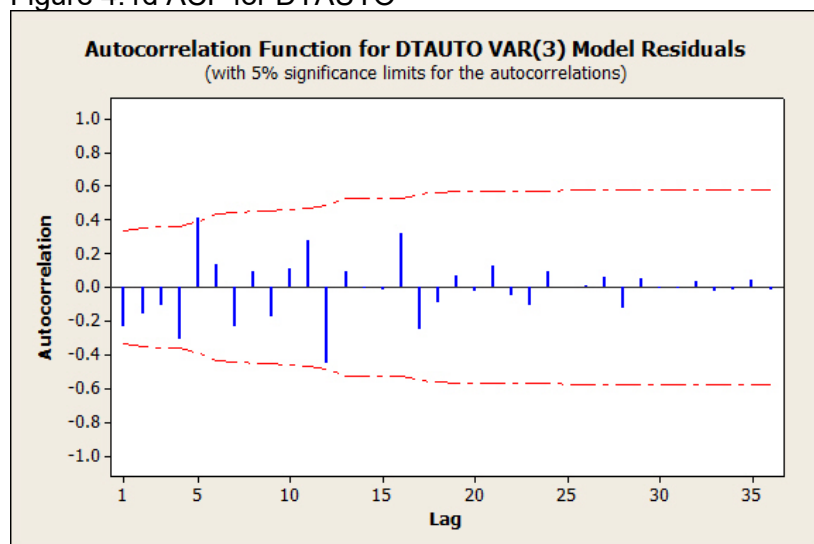


Figure 4.1e ACF for DTFURN

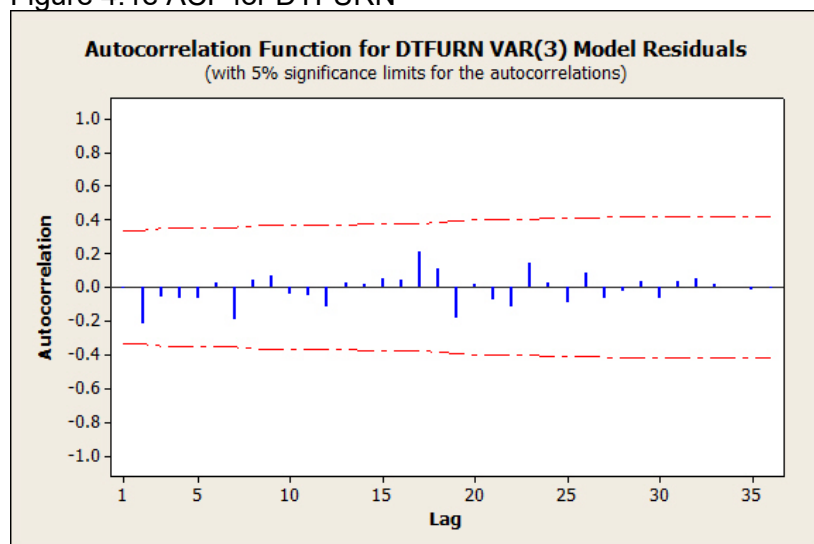
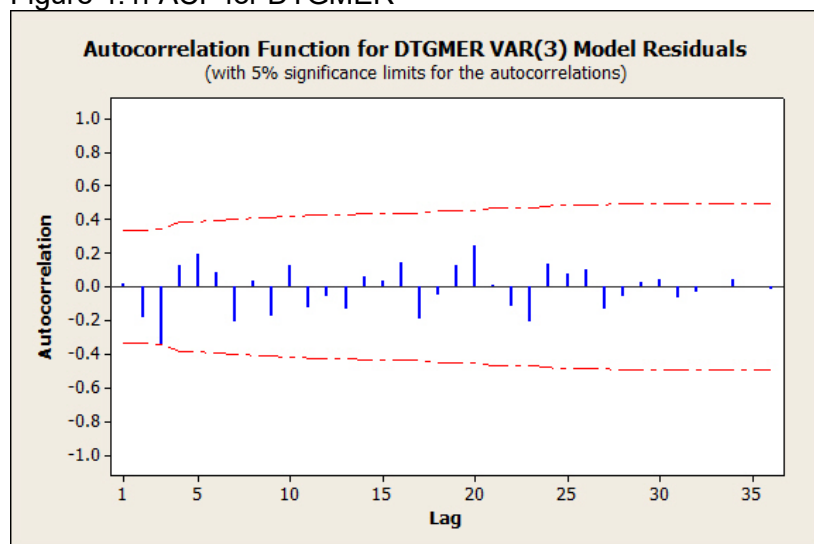


Figure 4.1f ACF for DTGMER



Next, the R prediction function was used to predict the next four values for each variable. The corresponding standard errors and prediction intervals had to be calculated manually since they are not

yet implemented in R for multivariate models. The ARMAto MA function was used to obtain the psi weights needed for calculating the standard errors. The VAR(3) model intercepts and variable coefficient estimates can be found in Table 4.1.

Table 4.1 VAR(3) intercepts and coefficient estimates

Lag 1 Variable Coefficients						
	dtwasa	dtempl	dtbldg	dtauto	dtfurn	dtgmer
dtwasa	0.96660	-0.00417	0.00392	-0.00065	0.01404	0.00259
dtempl	-3.38040	0.74246	0.38470	0.00913	0.20687	0.12561
dtbldg	1.02800	0.01192	1.09577	0.00301	-0.13090	-0.19307
dtauto	-29.66670	-1.95765	2.12978	-0.26242	0.17747	0.58931
dtfurn	-4.31810	0.35404	0.11443	-0.02920	0.82025	-0.18997
dtgmer	3.98340	0.08974	0.28743	-0.08503	-0.44333	0.42628
Lag 2 Variable Coefficients						
	dtwasa	dtempl	dtbldg	dtauto	dtfurn	dtgmer
dtwasa	-0.25050	-0.00069	-0.00302	0.00031	-0.00259	0.00215
dtempl	9.47390	-0.10105	-0.37860	-0.00795	0.28671	0.10910
dtbldg	11.65160	-0.08702	-0.18859	-0.04931	0.31174	0.34114
dtauto	-50.22990	0.84255	1.12377	-0.36062	3.01409	1.62749
dtfurn	9.28890	0.29382	0.05292	-0.09764	-0.14163	-0.02554
dtgmer	-8.60220	0.08913	-0.29630	0.16219	0.52138	0.29182
Lag 3 Variable Coefficients						
	dtwasa	dtempl	dtbldg	dtauto	dtfurn	dtgmer
dtwasa	-0.17160	0.00203	-0.00477	-0.00038	0.01582	-0.00293
dtempl	-10.72330	-0.08156	0.03298	0.00795	0.32724	-0.11656
dtbldg	-24.00310	0.36232	-0.33491	0.07507	-0.48325	-0.12557
dtauto	-128.86380	2.26683	-1.91969	0.06618	1.18607	0.92212
dtfurn	-19.45750	-0.14862	-0.24548	0.07737	-0.31639	-0.00157
dtgmer	20.54650	-1.18942	-0.18110	-0.07241	1.60431	-0.32118
Intercepts						
	dtwasa	dtempl	dtbldg	dtauto	dtfurn	dtgmer
	-4.591	-144.194	49.944	548.352	106.457	-202.640

5.0 – RESULTS

The prediction model's accuracy was measured by its ability to provide an interval in which the actual dealer sales falls. Overall, actual dealer sales were found to be within the prediction interval about 63% of the time. Individually, the model's accuracy was 25%, 100%, 50% and 75% for building material dealer sales, automotive dealer sales, furniture dealer sales and general merchandise dealer sales, respectively. A summary of the results is given in Table 5.1.

Table 5.1 VAR(3) model results

Sales (original scale)			Sales (detrended/differenced scale)		95 % Prediction Interval		
BLDG			Predicted	Actual	Lower Bound	Upper Bound	Good?
Q1-1990	17,271	19,368	-2,123.9	-26.6	-3,377.8	-869.9	FALSE
Q2-1990	26,972	26,220	-621.2	-1,373.6	-2,419.5	1,177.2	TRUE
Q3-1990	27,694	99,006	1,166.1	72,478.4	-1,076.7	3,408.9	FALSE
Q4-1990	26,397	87,063	2,636.7	63,302.4	-925.2	6,198.5	FALSE
AUTO			Predicted	Actual	Lower Bound	Upper Bound	Good?
Q1-1990	90,595	92,253	-3,241.3	-1,583.0	-7,899.6	1,416.9	TRUE
Q2-1990	113,333	103,038	4,259.1	-6,036.0	-134,014.5	142,532.7	TRUE
Q3-1990	113,094	22,620	4,859.1	-85,615.0	-4,088,150.1	4,097,868.2	TRUE
Q4-1990	96,639	25,611	6,001.8	-65,026.0	-121,139,471.9	121,151,475.5	TRUE
FURN			Predicted	Actual	Lower Bound	Upper Bound	Good?
Q1-1990	20,517	21,738	-3,343.9	-2,123.0	-4,164.0	-2,523.7	FALSE
Q2-1990	22,202	22,842	-1,672.0	-1,032.0	-5,307.4	1,963.3	TRUE
Q3-1990	22,939	49,137	-746.3	25,452.0	-16,748.2	15,255.6	FALSE
Q4-1990	27,996	71,265	885.7	44,155.0	-69,410.9	71,182.2	TRUE
GMER			Predicted	Actual	Lower Bound	Upper Bound	Good?
Q1-1990	39,367	41,832	658.8	3,124.0	-1,151.3	2,468.9	FALSE
Q2-1990	42,853	50,181	-6,120.9	1,207.0	-13,555.0	1,313.1	TRUE
Q3-1990	37,964	34,911	-11,234.0	-14,287.0	-41,059.5	18,591.5	TRUE
Q4-1990	60,729	36,543	-11,429.3	-35,615.0	-131,419.7	108,561.1	TRUE

6.0 - CONSIDERATIONS

As a rule of thumb, time series data should have at least 50 observations to make sure that there is enough information to model the data correctly. In this case we started with 44 observations and lost 4 due to seasonal differencing. So the model was fitted to series having only 40 observations. It is advisable that future studies have at least 50 observations.

Along with evaluating how well a particular model predicts dealer sales, the width of the prediction interval needs to be considered to make sure it is realistic for the contextual situation at hand. A prediction interval can be unrealistically wide or unrealistically narrow, thus it may not have a meaningful or useful interpretation.

I want to personally thank you for the opportunity to work on this project. Please don't hesitate to contact me if you have any further questions.

7.0 - RESOURCES

1: Basics of Time Series Models - <https://onlinecourses.science.psu.edu/stat510/node/41>

2: Seasonal Time Series Models - <https://onlinecourses.science.psu.edu/stat510/node/50>

3: Vector Autoregressive Models - <https://onlinecourses.science.psu.edu/stat510/node/79>

4: R function ar.ols - <http://stat.ethz.ch/R-manual/R-patched/library/stats/html/ar.ols.html>

5: Linear Trend Models - <http://people.duke.edu/~rnau/411trend.htm>

6.0 - Appendix

Data:

Table 1A Original data series

TIME	WASA	EMPL	BLDG	AUTO	FURN	GMER
1	1193.3	87973	9392	42960	9203	30363
2	1217.3	90021	13637	47516	10174	25812
3	1254.4	90120	14392	43888	11078	26026
4	1284.7	91180	13301	40298	12334	38081
5	1319.8	89832	10411	40016	10228	21909
6	1335.1	90668	13057	39073	10394	26615
7	1360.1	89850	13621	40484	11246	27039
8	1411.6	91276	13354	38703	13071	40170
9	1451.7	89964	11056	41919	10954	23641
10	1478.1	91566	14559	45398	11410	30477
11	1512.6	91405	14067	46635	11892	30065
12	1530.6	91691	12286	39970	13204	43805
13	1542.7	89341	9658	41554	10439	24256
14	1563.9	90310	13659	48384	11148	30976
15	1579.8	89324	12351	46041	11556	30579
16	1586	89436	12307	46411	13370	45471
17	1610.7	88815	10508	47558	11114	25891
18	1643.5	89920	16287	59342	12043	33050
19	1671.3	90480	16697	57487	13165	33385
20	1715.4	92367	15523	57165	15534	50311
21	1755.8	91637	13540	62200	13002	29177
22	1793.1	94256	19190	74678	14602	36855
23	1819.5	95021	18827	69074	15684	35418
24	1847.7	96547	17215	67030	19088	53126
25	1882.7	95450	13848	70453	15436	30325
26	1939.8	97657	20319	85102	16117	37977
27	1976	97985	20403	85142	16782	37028
28	2012.8	99383	19179	71203	20491	54369
29	2044.1	98213	16790	73380	16383	31594
30	2069.8	99850	24400	88139	18180	37617
31	2097.9	99877	24215	92899	19782	36297
32	2128.4	101169	22042	81838	23605	52457
33	2163.2	99922	18182	75122	18483	30718
34	2211.9	102186	22670	90515	19659	41090
35	2265.1	102657	21204	89907	21042	40706
36	2325	104522	19746	77163	24203	60491
37	2358.7	103445	16656	84508	20242	34723
38	2405.6	106050	24552	99279	22143	42871
39	2452.2	105963	23775	94556	22870	42072
40	2505.1	107644	22449	87039	27486	64101
41	2560.7	106425	18145	87696	22399	36516

42	2572.9	108741	26344	102934	22412	46782
43	2586.6	108897	25278	102095	22223	47006
44	2612.7	110260	22511	84497	25648	69966

Table A2 Actual values for 1990

Values for 1990:

Quarter	BDLG	AUTO	FURN	GMER
1	19368	92253	21738	41832
2	26220	103038	22842	50181
3	99006	22620	49137	34911
4	87063	25611	71265	36543

Table A3 Standard errors for each predicted quarter in 1990

Quarter	dtbldg	dtauto	dtfurn	dtgmer
1	639.78	2,376.66	418.46	923.52
2	917.54	70,547.74	1,854.77	3,792.90
3	1,144.28	2,088,269.96	8,164.23	15,217.09
4	1,817.28	61,808,915.17	35,865.57	61,219.58

R Program Code and Output:

```
> setwd("C:/Users/Shenee/Documents/STAT581")
>
> sales <- read.csv("sales.csv")
> attach(sales)
> source("itall.R")
itall has been installed
> library(car)
>
> # Designate data as a time series
>
> wasa = ts(wasa)
> empl = ts(empl)
> bldg = ts(bldg)
> auto = ts(auto)
> furn = ts(furn)
> gmer = ts(gmer)
>
> # Linear regressions between
> # each variable and time (de-trending)
>
> wasa.lm = lm (wasas~time(wasas))
> empl.lm = lm (empl~time(empl))
> bldg.lm = lm (bldg~time(bldg))
> auto.lm = lm (auto~time(auto))
> furn.lm = lm (furn~time(furn))
> gmer.lm = lm (gmer~time(gmer))
>
> wasa.lm
```

Call:

lm(formula = wasa ~ time(wasa))

Coefficients:

(Intercept)	time(wasa)
1109.48	33.16

> empl.lm

Call:

lm(formula = empl ~ time(empl))

Coefficients:

(Intercept)	time(empl)
85206.6	495.2

> bldg.lm

Call:

lm(formula = bldg ~ time(bldg))

Coefficients:

(Intercept)	time(bldg)
10053.3	312.4

> auto.lm

Call:

lm(formula = auto ~ time(auto))

Coefficients:

(Intercept)	time(auto)
31531	1535

> furn.lm

Call:

lm(formula = furn ~ time(furn))

Coefficients:

(Intercept)	time(furn)
7948.0	365.5

> gmer.lm

Call:

lm(formula = gmer ~ time(gmer))

Coefficients:

(Intercept)	time(gmer)
25562	548

>

> wasa.res = residuals(wasa.lm)

> empl.res = residuals(empl.lm)

> bldg.res = residuals(bldg.lm)

> auto.res = residuals(auto.lm)

> furn.res = residuals(furn.lm)

```

> gmer.res = residuals(gmer.lm)
>
> # Exploratory Data Analysis
>
> par(mfrow=c(3,1)) # plot time series
> plot(wasa, type="o", xlab=NULL, main="wasa")
> plot(wasa.res, type="o", xlab="Time", main="detrended")
> plot(diff(wasa.res,4), type="o", xlab="Time", main="fourth difference(detrended)")
>
> plot(empl, type="o", xlab=NULL, main="empl")
> plot(empl.res, type="o", xlab=NULL, main="detrended")
> plot(diff(empl.res,4), type="o", xlab="Time", main="fourth difference(detrended)")
>
> plot(bldg, type="o", xlab=NULL, main="bldg")
> plot(bldg.res, type="o", xlab=NULL, main="detrended")
> plot(diff(bldg.res,4), type="o", xlab="Time", main="fourth difference(detrended)")
>
> plot(auto, type="o", xlab=NULL, main="auto")
> plot(auto.res, type="o", xlab=NULL, main="detrended")
> plot(diff(auto.res,4), type="o", xlab="Time", main="fourth difference(detrended)")
>
> plot(furn, type="o", xlab=NULL, main="furn")
> plot(furn.res, type="o", xlab=NULL, main="detrended")
> plot(diff(furn.res,4), type="o", xlab="Time", main="fourth difference(detrended)")
>
> plot(gmer, type="o", xlab=NULL, main="gmer")
> plot(gmer.res, type="o", xlab=NULL, main="detrended")
> plot(diff(gmer.res,4), type="o", xlab="Time", main="fourth difference(detrended)")
>
> # Identify AR order
>
> par(mfrow=c(3,1)) # plot ACFs
> acf(wasa, 40, xlab="", main="wasa")
> acf(wasa.res,40, xlab="", main="detrended")
> acf(diff(wasa.res,4),40, main="fourth difference") #AR(3)
>
> acf(empl, 40, xlab="", main="empl")
> acf(empl.res,40, xlab="", main="detrended")
> acf(diff(empl.res,4),40, main="fourth difference") #AR(3)
>
> acf(bldg, 40, xlab="", main="bldg")
> acf(bldg.res,40, xlab="", main="detrended")
> acf(diff(bldg.res,4),40, main="fourth difference") #AR(3)
>
> acf(auto, 40, xlab="", main="auto")
> acf(auto.res,40, xlab="", main="detrended")
> acf(diff(auto.res,4),40, main="fourth difference") #AR(3)
>
> acf(furn, 40, xlab="", main="furn")
> acf(furn.res,40, xlab="", main="detrended")
> acf(diff(furn.res,4),40, main="fourth difference") #AR(3)
>
> acf(gmer, 40, xlab="", main="gmer")
> acf(gmer.res,40, xlab="", main="detrended")
> acf(diff(gmer.res,4),40, main="fourth difference") #AR(2)
>

```

```

> # Store the differenced residuals from each regression
>
> dtwasa = diff(residuals( wasa.lm),4)
> dtempl = diff(residuals( empl.lm),4)
> dtbldg = diff(residuals( bldg.lm),4)
> dtauto = diff(residuals( auto.lm),4)
> dtfurn = diff(residuals( furn.lm),4)
> dtgmer = diff(residuals( gmer.lm),4)
>
> dtwasa
      5      6      7      8      9     10
-6.1576744 -14.8576744 -26.9576744 -5.7576744 -0.7576744 10.3423256
     11     12     13     14     15     16
19.8423256 -13.6576744 -41.6576744 -46.8576744 -65.4576744 -77.2576744
     17     18     19     20     21     22
-64.6576744 -53.0576744 -41.1576744 -3.2576744 12.4423256 16.9423256
     23     24     25     26     27     28
15.5423256 -0.3576744 -5.7576744 14.0423256 23.8423256 32.4423256
     29     30     31     32     33     34
28.7423256 -2.6576744 -10.7576744 -17.0576744 -13.5576744  9.4423256
     35     36     37     38     39     40
34.5423256 63.9423256 62.8423256 61.0423256 54.4423256 47.4423256
     41     42     43     44
69.3423256 34.6423256  1.7423256 -25.0576744
> dtempl
      5      6      7      8      9     10
-121.63312 -1333.63312 -2250.63312 -1884.63312 -1848.63312 -1082.63312
     11     12     13     14     15     16
-425.63312 -1565.63312 -2603.63312 -3236.63312 -4061.63312 -4235.63312
     17     18     19     20     21     22
-2506.63312 -2370.63312 -824.63312  950.36688  841.36688 2355.36688
     23     24     25     26     27     28
2560.36688 2199.36688 1832.36688 1420.36688  983.36688  855.36688
     29     30     31     32     33     34
 782.36688 212.36688 -88.63312 -194.63312 -271.63312  355.36688
     35     36     37     38     39     40
 799.36688 1372.36688 1542.36688 1883.36688 1325.36688 1141.36688
     41     42     43     44
 999.36688 710.36688 953.36688 635.36688
> dtbldg
      5      6      7      8      9     10     11     12
-230.518 -1829.518 -2020.518 -1196.518 -604.518  252.482 -803.518 -2317.518
     13     14     15     16     17     18     19     20
-2647.518 -2149.518 -2965.518 -1228.518 -399.518 1378.482 3096.482 1966.482
     21     22     23     24     25     26     27     28
1782.482 1653.482  880.482  442.482 -941.518 -120.518  326.482  714.482
     29     30     31     32     33     34     35     36
1692.482 2831.482 2562.482 1613.482 142.482 -2979.518 -4260.518 -3545.518
     37     38     39     40     41     42     43     44
-2775.518 632.482 1321.482 1453.482 239.482  542.482  253.482 -1187.518
> dtauto
      5      6      7      8      9     10
-9084.93672 -14583.93672 -9544.93672 -7735.93672 -4237.93672  184.06328
     11     12     13     14     15     16
 10.06328 -4873.93672 -6505.93672 -3154.93672 -6734.93672  300.06328
     17     18     19     20     21     22

```

```

-136.93672 4817.06328 5305.06328 4613.06328 8501.06328 9195.06328
  23      24      25      26      27      28
5446.06328 3724.06328 2112.06328 4283.06328 9927.06328 -1967.93672
  29      30      31      32      33      34
-3213.93672 -3103.93672 1616.06328 4494.06328 -4398.93672 -3764.93672
  35      36      37      38      39      40
-9132.93672 -10815.93672 3245.06328 2623.06328 -1491.93672 3735.06328
  41      42      43      44
-2952.93672 -2485.93672 1398.06328 -8682.93672
> dtfurn
  5      6      7      8      9      10
-436.83932 -1241.83932 -1293.83932 -724.83932 -735.83932 -445.83932
  11      12      13      14      15      16
-815.83932 -1328.83932 -1976.83932 -1723.83932 -1797.83932 -1295.83932
  17      18      19      20      21      22
-786.83932 -566.83932 147.16068 702.16068 426.16068 1097.16068
  23      24      25      26      27      28
1057.16068 2092.16068 972.16068 53.16068 -363.83932 -58.83932
  29      30      31      32      33      34
-514.83932 601.16068 1538.16068 1652.16068 638.16068 17.16068
  35      36      37      38      39      40
-201.83932 -863.83932 297.16068 1022.16068 366.16068 1821.16068
  41      42      43      44
695.16068 -1192.83932 -2108.83932 -3299.83932
> dtgmer
  5      6      7      8      9      10
-10645.8658 -1388.8658 -1178.8658 -102.8658 -459.8658 1670.1342
  11      12      13      14      15      16
834.1342 1443.1342 -1576.8658 -1692.8658 -1677.8658 -525.8658
  17      18      19      20      21      22
-556.8658 -117.8658 614.1342 2648.1342 1094.1342 1613.1342
  23      24      25      26      27      28
-158.8658 623.1342 -1043.8658 -1069.8658 -581.8658 -948.8658
  29      30      31      32      33      34
-922.8658 -2551.8658 -2922.8658 -4103.8658 -3067.8658 1281.1342
  35      36      37      38      39      40
2217.1342 5842.1342 1813.1342 -410.8658 -825.8658 1418.1342
  41      42      43      44
-398.8658 1719.1342 2742.1342 3673.1342
>
> # Check for multicollinearity
>
> vif(lm(dtwasa ~ dtempl + dtbldg + dtauto + dtfurn + dtgmer))
dtempl dtbldg dtauto dtfurn dtgmer
1.825063 1.975895 2.130466 2.332130 1.455447
> vif(lm(dtempl ~ dtwasa + dtbldg + dtauto + dtfurn + dtgmer))
dtwasa dtbldg dtauto dtfurn dtgmer
1.549215 1.950257 2.267401 2.296400 1.481540
> vif(lm(dtbldg ~ dtwasa + dtempl + dtauto + dtfurn + dtgmer))
dtwasa dtempl dtauto dtfurn dtgmer
3.083398 3.585287 1.826056 2.382238 1.400384
> vif(lm(dtauto ~ dtwasa + dtempl + dtbldg + dtfurn + dtgmer))
dtwasa dtempl dtbldg dtfurn dtgmer
2.857248 3.582350 1.569357 2.052312 1.292780
> vif(lm(dtfurn ~ dtwasa + dtempl + dtbldg + dtauto + dtgmer))
dtwasa dtempl dtbldg dtauto dtgmer

```

```

2.997556 3.477189 1.962157 1.966910 1.270025
> vif(lm(dtgmer ~ dtwasa + dtempl + dtbldg + dtauto + dtfurn))
  dtwasa dtempl dtbldg dtauto dtfurn
2.995226 3.591804 1.846776 1.983738 2.033437
>
> # Estimation of AR(1) model
>
> fitvar1 = ar.ols(cbind(dtwasa, dtempl, dtbldg, dtauto, dtfurn, dtgmer), order = 1)
> fitvar1

```

Call:
ar.ols(x = cbind(dtwasa, dtempl, dtbldg, dtauto, dtfurn, dtgmer), order.max = 1)

```

$ar
, , 1

```

```

      dtwasa dtempl dtbldg dtauto dtfurn dtgmer
dtwasa  0.7712 -0.001431 -0.002945 3.367e-05 0.01182 0.001695
dtempl -0.7412  0.744412  0.063565 2.379e-02 0.21190 0.137747
dtbldg 13.8847 -0.324752  0.778510 7.232e-02 -0.12385 0.025189
dtauto -31.9814 -0.130278  0.786198 3.181e-01 1.04854 0.920572
dtfurn -0.7856 -0.061745  0.091607 -3.407e-03 0.86440 0.050824
dtgmer -0.5963  0.315851 -0.026267 -3.016e-02 -0.50394 0.358828

```

```

$x.intercept
  dtwasa dtempl dtbldg dtauto dtfurn dtgmer
-1.054  5.729 -36.569 151.550 -59.904 356.421

```

```

$var.pred
      dtwasa dtempl dtbldg dtauto dtfurn dtgmer
dtwasa 228.7  5614   969.6   457.5  2036  10310
dtempl 5614.0 336044  92573.1 292187.8 112506 524408
dtbldg  969.6  92573 1155505.3 2418756.5 364828 -452689
dtauto  457.5 292188 2418756.5 16241773.8 1416701 -1010291
dtfurn 2035.9 112506 364828.4 1416700.6 531512 -234677
dtgmer 10309.5 524408 -452688.9 -1010291.2 -234677 2411351

```

```

> fitvar1$asy.se.coef
$x.mean
  dtwasa dtempl dtbldg dtauto dtfurn dtgmer
2.448375 93.847200 174.024068 652.439254 118.026591 251.393220

```

```

$ar
      dtwasa dtempl dtbldg dtauto dtfurn dtgmer
dtwasa 0.1275522 0.003092099 0.001897585 0.0006543504 0.003577312 0.001183212
dtempl 4.8891262 0.118521403 0.072735221 0.0250815185 0.137119807 0.045353013
dtbldg 9.0660737 0.219778285 0.134875404 0.0465095163 0.254265941 0.084099641
dtauto 33.9899099 0.823977868 0.505665733 0.1743703302 0.953276653 0.315300681
dtfurn 6.1487919 0.149058013 0.091475187 0.0315436809 0.172448229 0.057038053
dtgmer 13.0967486 0.317489251 0.194839498 0.0671871267 0.367309732 0.121489400

```

```

>
>
> # Estimation of AR(2) model
>

```

```
> fitvar2 = ar.ols(cbind(dtwasa, dtempl, dtbldg, dtauto, dtfurn, dtgmer), order = 2)
> fitvar2
```

Call:

```
ar.ols(x = cbind(dtwasa, dtempl, dtbldg, dtauto, dtfurn, dtgmer), order.max = 2)
```

\$ar

, , 1

	dtwasa	dtempl	dtbldg	dtauto	dtfurn	dtgmer
dtwasa	1.065	0.002615	0.0007558	0.0002732	0.008674	0.001617
dtempl	2.236	0.747502	0.2695238	0.0216795	0.052585	0.190852
dtbldg	18.336	-0.190858	1.2031760	-0.0154303	-0.095731	-0.163346
dtauto	-4.641	-1.469119	2.7949583	0.0157934	0.051973	0.868780
dtfurn	7.871	-0.021926	0.2960990	-0.0344243	0.884180	-0.053035
dtgmer	9.038	0.589136	-0.0591857	-0.0369488	-0.945354	0.479738

, , 2

	dtwasa	dtempl	dtbldg	dtauto	dtfurn	dtgmer
dtwasa	-0.4356	-0.0003585	-0.003819	-0.0008876	0.002507	-0.001066
dtempl	-10.1300	0.0088864	-0.191580	-0.0385121	0.456170	0.028317
dtbldg	-25.2543	0.4382579	-0.397645	-0.0233934	-0.250249	0.190775
dtauto	-111.6694	2.1989813	-1.710806	-0.0905430	2.527310	0.845756
dtfurn	-21.0715	0.3692969	-0.143522	-0.0386703	-0.299467	0.077411
dtgmer	-2.7505	-0.9210847	0.042224	0.0275541	1.190246	0.065375

\$x.intercept

	dtwasa	dtempl	dtbldg	dtauto	dtfurn	dtgmer
	-1.499	-47.965	148.256	346.890	56.609	135.118

\$var.pred

	dtwasa	dtempl	dtbldg	dtauto	dtfurn	dtgmer
dtwasa	156.1	3575	-842	-8007	452.7	7621
dtempl	3574.6	244431	35123	-240001	81035.4	371674
dtbldg	-842.0	35123	727972	1231668	128100.2	-205980
dtauto	-8007.3	-240001	1231668	10283077	876238.9	-1304031
dtfurn	452.7	81035	128100	876239	387512.5	-54388
dtgmer	7621.1	371674	-205980	-1304031	-54387.9	1513788

```
> fitvar2$asy.se.coef
```

\$x.mean

	dtwasa	dtempl	dtbldg	dtauto	dtfurn	dtgmer
	2.229504	88.214113	152.235930	572.165162	111.071516	219.529238

\$ar

, , dtwasa

	dtwasa	dtempl	dtbldg	dtauto	dtfurn	dtgmer
1	0.1758024	6.955922	12.00422	45.11678	8.758291	17.31048
2	0.1795620	7.104677	12.26093	46.08161	8.945590	17.68067

, , dtempl

	dtwasa	dtempl	dtbldg	dtauto	dtfurn	dtgmer
--	--------	--------	--------	--------	--------	--------

```
1 0.005590876 0.2212125 0.3817586 1.434805 0.2785315 0.5505084
2 0.005065670 0.2004318 0.3458962 1.300020 0.2523663 0.4987938
```

```
, , dtbldg
```

```
      dtwasa  dtempl  dtbldg  dtauto  dtfurn  dtgmer
1 0.002443344 0.09667505 0.1668374 0.6270436 0.1217248 0.2405851
2 0.002402917 0.09507550 0.1640770 0.6166687 0.1197108 0.2366044
```

```
, , dtauto
```

```
      dtwasa  dtempl  dtbldg  dtauto  dtfurn  dtgmer
1 0.0006764696 0.02676567 0.04619099 0.1736047 0.03370100 0.06660893
2 0.0005968058 0.02361363 0.04075134 0.1531603 0.02973223 0.05876478
```

```
, , dtfurn
```

```
      dtwasa  dtempl  dtbldg  dtauto  dtfurn  dtgmer
1 0.003646714 0.1442885 0.2490065 0.9358687 0.1816754 0.3590756
2 0.004554286 0.1801981 0.3109777 1.1687819 0.2268897 0.4484401
```

```
, , dtgmer
```

```
      dtwasa  dtempl  dtbldg  dtauto  dtfurn  dtgmer
1 0.001893200 0.07490768 0.12927230 0.4858584 0.09431722 0.1864149
2 0.001274862 0.05044208 0.08705066 0.3271722 0.06351227 0.1255299
```

```
>
> # Estimation of AR(3) model
>
> fitvar3 = ar.ols(cbind(dtwasa, dtempl, dtbldg, dtauto, dtfurn, dtgmer), order = 3)
> fitvar3
```

```
Call:
```

```
ar.ols(x = cbind(dtwasa, dtempl, dtbldg, dtauto, dtfurn, dtgmer), order.max = 3)
```

```
$ar
```

```
, , 1
```

```
      dtwasa  dtempl  dtbldg  dtauto  dtfurn  dtgmer
dtwasa  0.9666 -0.004172 0.00392 -0.0006546 0.01404 0.002592
dtempl -3.3804 0.742456 0.38470 0.0091283 0.20687 0.125614
dtbldg  1.0280 0.011918 1.09577 0.0030133 -0.13090 -0.193069
dtauto -29.6667 -1.957646 2.12978 -0.2624200 0.17747 0.589305
dtfurn -4.3181 0.354035 0.11443 -0.0291982 0.82025 -0.189970
dtgmer  3.9834 0.089735 0.28743 -0.0850336 -0.44333 0.426277
```

```
, , 2
```

```
      dtwasa  dtempl  dtbldg  dtauto  dtfurn  dtgmer
dtwasa -0.2505 -0.0006924 -0.003017 0.0003092 -0.00259 0.00215
dtempl  9.4739 -0.1010549 -0.378600 -0.0079514 0.28671 0.10910
dtbldg 11.6516 -0.0870150 -0.188585 -0.0493123 0.31174 0.34114
dtauto -50.2299 0.8425495 1.123772 -0.3606237 3.01409 1.62749
dtfurn  9.2889 0.2938193 0.052922 -0.0976428 -0.14163 -0.02554
```



```
dtgmer -8.6022 0.0891250 -0.296304 0.1621860 0.52138 0.29182
```

```
, , 3
```

```
dtwasa dttempl dtbldg dtauto dtfurn dtgmer
dtwasa -0.1716 0.002027 -0.004772 -0.000376 0.01582 -0.002925
dttempl -10.7233 -0.081562 0.032982 0.007946 0.32724 -0.116559
dtbldg -24.0031 0.362319 -0.334908 0.075072 -0.48325 -0.125571
dtauto -128.8638 2.266825 -1.919693 0.066184 1.18607 0.922123
dtfurn -19.4575 -0.148618 -0.245479 0.077368 -0.31639 -0.001573
dtgmer 20.5465 -1.189421 -0.181099 -0.072409 1.60431 -0.321184
```

```
$x.intercept
```

```
dtwasa dttempl dtbldg dtauto dtfurn dtgmer
-4.591 -144.194 49.944 548.352 106.457 -202.640
```

```
$var.pred
```

```
dtwasa dttempl dtbldg dtauto dtfurn dtgmer
dtwasa 83.53 1509 -767.1 -4344 878.8 2525
dttempl 1508.87 143295 3334.5 -49026 59996.9 195861
dtbldg -767.08 3335 409321.7 574999 -66687.4 -134037
dtauto -4344.06 -49026 574999.4 5648519 335524.7 -550317
dtfurn 878.82 59997 -66687.4 335525 175108.9 20137
dtgmer 2525.40 195861 -134036.9 -550317 20136.9 852891
```

```
> fitvar3$asy.se.coef
```

```
$x.mean
```

```
dtwasa dttempl dtbldg dtauto dtfurn dtgmer
1.877095 77.748654 131.404469 488.140722 85.947238 189.681257
```

```
$ar
```

```
, , dtwasa
```

```
dtwasa dttempl dtbldg dtauto dtfurn dtgmer
1 0.1513943 6.270699 10.59823 39.37024 6.931943 15.29845
2 0.2105197 8.719656 14.73726 54.74589 9.639142 21.27311
3 0.1577051 6.532091 11.04001 41.01138 7.220899 15.93616
```

```
, , dttempl
```

```
dtwasa dttempl dtbldg dtauto dtfurn dtgmer
1 0.004655401 0.1928251 0.3258974 1.210642 0.2131585 0.4704301
2 0.005245686 0.2172745 0.3672198 1.364146 0.2401861 0.5300787
3 0.004054214 0.1679242 0.2838118 1.054303 0.1856317 0.4096799
```

```
, , dtbldg
```

```
dtwasa dttempl dtbldg dtauto dtfurn dtgmer
1 0.002028547 0.08402174 0.1420067 0.5275259 0.09288182 0.2049855
2 0.002962453 0.12270378 0.2073840 0.7703891 0.13564288 0.2993570
3 0.002131748 0.08829628 0.1492312 0.5543634 0.09760711 0.2154140
```

```
, , dtauto
```

```
dtwasa dttempl dtbldg dtauto dtfurn dtgmer
```

```

1 0.0006079436 0.02518082 0.04255858 0.1580964 0.02783613 0.06143295
2 0.0005401103 0.02237119 0.03780997 0.1404563 0.02473023 0.05457837
3 0.0005186286 0.02148142 0.03630615 0.1348699 0.02374663 0.05240763

```

```
, , dtfurn
```

```

      dtwasa  dtempl  dtbldg  dtauto  dtfurn  dtgmer
1 0.003139484 0.1300364 0.2197769 0.8164263 0.1437487 0.3172461
2 0.004141614 0.1715442 0.2899302 1.0770312 0.1896335 0.4185118
3 0.004847752 0.2007922 0.3393628 1.2606632 0.2219658 0.4898673

```

```
, , dtgmer
```

```

      dtwasa  dtempl  dtbldg  dtauto  dtfurn  dtgmer
1 0.001744058 0.07223830 0.12209132 0.4535443 0.07985582 0.1762378
2 0.001862479 0.07714327 0.13038129 0.4843398 0.08527801 0.1882043
3 0.001081424 0.04479222 0.07570418 0.2812255 0.04951556 0.1092784

```

```

>
>
>
> # Create ACFs for the residuals each variable
> # in the AR(1) model
>
> acf(fitvar1$resid[,1], 40, na.action = na.omit)
> acf(fitvar1$resid[,2], 40, na.action = na.omit)
> acf(fitvar1$resid[,3], 40, na.action = na.omit)
> acf(fitvar1$resid[,4], 40, na.action = na.omit)
> acf(fitvar1$resid[,5], 40, na.action = na.omit)
> acf(fitvar1$resid[,6], 40, na.action = na.omit)
>
> # Create ACFs for the residuals each variable
> # in the AR(2) model
>
> acf(fitvar2$resid[,1], 40, na.action = na.omit)
> acf(fitvar2$resid[,2], 40, na.action = na.omit)
> acf(fitvar2$resid[,3], 40, na.action = na.omit)
> acf(fitvar2$resid[,4], 40, na.action = na.omit)
> acf(fitvar2$resid[,5], 40, na.action = na.omit)
> acf(fitvar2$resid[,6], 40, na.action = na.omit)
>
> # Create ACFs for the residuals each variable
> # in the AR(3) model
>
> acf(fitvar3$resid[,1], 40, na.action = na.omit)
> acf(fitvar3$resid[,2], 40, na.action = na.omit)
> acf(fitvar3$resid[,3], 40, na.action = na.omit)
> acf(fitvar3$resid[,4], 40, na.action = na.omit)
> acf(fitvar3$resid[,5], 40, na.action = na.omit)
> acf(fitvar3$resid[,6], 40, na.action = na.omit)
>
> predict(fitvar3, n.ahead = 4, level=0.95)
$pred
Time Series:
Start = 41

```

```

End = 44
Frequency = 1
  dtwasa dtmpl dtbldg dtauto dtfurn dtgmer
41 -82.2023 -1679.930 -2123.8544 -3241.347 -3343.8583 658.8342
42 -141.6203 -3804.669 -621.1629 4259.091 -1672.0230 -6120.9485
43 -176.1889 -5690.790 1166.1401 4859.065 -746.3367 -11234.0390
44 -184.4836 -6818.199 2636.6847 6001.793 885.6571 -11429.3017

```

\$se

Time Series:

Start = 41

End = 44

Frequency = 1

	Series 1	Series 2	Series 3	Series 4	Series 5	Series 6
41	NA	NA	NA	NA	NA	NA
42	NA	NA	NA	NA	NA	NA
43	NA	NA	NA	NA	NA	NA
44	NA	NA	NA	NA	NA	NA

Warning message:

In predict.ar(fitvar3, n.ahead = 4, level = 0.95) :

'se.fit' not yet implemented for multivariate models

```

>
> w.psi = ARMAtoMA(ar = list(0.9666, -0.004172, 0.00392, -0.0006546, 0.01404, 0.002592, -0.2505, -
0.0006924, -0.003017, 0.0003092, -0.00259, 0.00215, -0.1716, 0.002027, -0.004772, -0.000376,
0.01582, -0.002925), ma = 0, 4)
> e.psi = ARMAtoMA(ar = list(-3.3804, 0.742456, 0.3847, 0.0091283, 0.20687, 0.125614, 9.4739, -
0.1010549, -0.3786, -0.0079514, 0.28671, 0.1091, -10.7233, -0.081562, 0.032982, 0.007946, 0.32724, -
-0.116559), ma = 0, 4)
> b.psi = ARMAtoMA(ar = list(1.028, 0.011918, 1.09577, 0.0030133, -0.1309, -0.193069, 11.6516, -
0.087015, -0.188585, -0.0493123, 0.31174, 0.34114, -24.0031, 0.362319, -0.334908, 0.075072, -
0.48325, -0.125571), ma = 0, 4)
> a.psi = ARMAtoMA(ar = list(-29.6667, -1.957646, 2.12978, -0.26242, 0.17747, 0.589305, -50.2299,
0.8425495, 1.123772, -0.3606237, 3.01409, 1.62749, -128.8638, 2.266825, -1.919693, 0.066184,
1.18607, 0.922123), ma = 0, 4)
> f.psi = ARMAtoMA(ar = list(-4.3181, 0.354035, 0.11443, -0.0291982, 0.82025, -0.18997, 9.2889,
0.2938193, 0.052922, -0.0976428, -0.14163, -0.02554, -19.4575, -0.148618, -0.245479, 0.077368, -
0.31639, -0.001573), ma = 0, 4)
> g.psi = ARMAtoMA(ar = list(3.9834, 0.089735, 0.28743, -0.0850336, -0.44333, 0.426277, -8.6022,
0.089125, -0.296304, 0.162186, 0.52138, 0.29182, 20.5465, -1.189421, -0.181099, -0.072409,
1.60431, -0.321184), ma = 0, 4)
>
> w.psi
[1] 0.9666000 0.9301436 0.8989641 0.8681926
> e.psi
[1] -3.38040 12.16956 -43.26308 153.99057
> b.psi
[1] 1.028000 1.068702 2.206647 3.410635
> a.psi
[1] -29.6667 878.1554 -25991.7674 769307.4025
> f.psi
[1] -4.31810 19.00002 -83.45833 366.58475
> g.psi
[1] 3.98340 15.95721 64.20883 258.26130
>

```